BHARATIYA ANTARIKSH STATION

AIM: Establish India's space station for microgravity research

TARGET: 2035

PRECURSORS

- Sending five modules by as many launchers; first in orbit by 2028
- Docking of the modules to build the space station



CREWED LUNAR MISSION

AIM: Land first Indian astronaut on the moon

TARGET: 2040

PRECURSORS

- Chandrayaan-4, with expected launch in 2027-28, to bring back lunar samples
- Chandrayaan-5, a joint mission with Japan, to study the moon's potential to host humans





"From a follower, India intends to be a frontliner in space. India is now an equal partner in global space endeavours"

—JITENDRA SINGH
Union MoS for science & technology, earth sciences and space

aamras to share with fellow astronauts. Fortunately, as Godard points out, microgravity has no bearing on the peristaltic movement (muscle contraction and relaxation) of food down the digestive tract. Meanwhile, Shukla is busy absorbing the celestial joy of witnessing 16 sunrises and sunsets daily at the ISS as the spacecraft's speed of 28,000 kmph means it orbits the Earth once every 90 minutes.

Shukla's observations, therefore, will prove valuable not only in understanding some of the challenges faced aboard the ISS but also the rigorous protocols to be observed for a human spaceflight. ISRO is taking no chances with its first crew spaceflight: it is planning three unmanned tests that will fully validate all the critical systems (including a robot called Vyommitra on board as a substitute) and ensure they are all error-free before it attempts one with the selected astronauts aboard.

Shukla's experience will also be crucial in the construction of India's first space station, the BAS-1. While the first module of BAS-1 was sanctioned along with the Gaganyaan project, the ultimate aim is to have five modules in space, including the living quarters for astronauts and orbiting laboratories. Already, the technology for docking spacecraft with BAS-1 has been validated with the Space Docking Experiment (SpaDeX) earlier this year, using two satellites as a demonstration. Autonomous rendezvous apart from docking and undocking capabilities were validated and these would be used for future space endeavours, including in BAS-1. Of course, the technology needs to be human-rated and would require many more SpaDeX tests. "It will not be like a five-star hotel," says